

COURSE DESCRIPTION

Course Number	0418-427	Course Title	Computer Security & Privacy الخصوصية و أمن الكمبيوتر
Credit Hours	3 (3:1:3)	Course Coordinator	Hussain Almohri (Systems Track)
Compulsory	No	Prerequisite	0418-321, 0418-331

Course Description

This course introduces the students to the principles of computer security, techniques and models. The course includes classical attack scenarios, defensive resources and techniques, and models for protecting systems and networks.

تقدم هذه المادة مبادئ وتقنيات ونماذج أمن الكمبيوتر. وتتضمن المادة السيناريوهات الهجومية الكلاسيكية، والموارد والتقنيات الدفاعية، ونماذج لحماية الأنظمة والشبكات.

Textbook

Title: Introduction to Computer Security. 1st ed.
Author: Goodrich, Michael, and Roberto Tamassia.
Publisher: Pearson, 2011. ISBN-10: 9780321512949.

Major Topics Covered in the Course

	Topic	Hours
1	Fundamental concepts: confidentiality, integrity, availability, basic symmetric and asymmetric encryption functions, Saltzer and Schroeder principles	6
2	Authentication and Authorization: basic authentication systems, cryptographic protocols, authentication systems such as Kerberos, basic access control models	6
3	Malware: taxonomy, historical attacks, countermeasures	3

4	Operating system security: security of kernel, process security, address space randomization, application program integrity, sandboxing, secure coding practices, example systems (such as SELinux)	6
5	Cryptography: Basic encryption and cipher algorithms, strong cryptographic keys, symmetric and asymmetric encryption/decryption algorithms, hashing properties and functions, public key infrastructure, digital signature, key exchange protocol attacks on cryptographic primitives	6
6	Network security: fundamentals, protocols, denial of service, DNSSEC, firewalls, intrusion detection, wireless security	9
7	Security of the Web: attacks, defenses, and secure programming methods for the Web	3
8	Exams	3
	Total	42

Lab Hours

	Topics	Hours
1	Example attacks	2
2	Secure Programming (memory bugs, vulnerable functions, heap overflow, return-to-libc, writing safe string libraries)	4
3	Security Configuration for Operating Systems (file system access control, SELinux)	2
4	Cryptography (writing programs using crypto libraries such as openssl)	3
5	Network Security Tools (penetration testing, traffic analysis)	2
6	Web security	1
	Total	14

Grading

Midterm Exam(s)	30%
Lab	10%
Project (s)	20%
Final Exam	40%

Learning Outcomes

1. Understand and analyze security concepts, attacks, defenses, and secure protocols.
2. Analyze security problems.
3. Develop security solutions.

Relationship with Student Outcomes

Course Outcomes	Student Outcomes					
	1	2	3	4	5	6
1	M		M	M		
2	H					
3		H				H

Last updated on March 16, 2021 by Hussain Almohri

Approved by the curricula committee of the CS department on April, 1, 2021